

Manisha Lagisetty

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EDUCATION

Master of Science in Data Science , San Jose State University, San Jose, California	Jan 2023 – Dec 2024
Bachelor of Computer Science and Engineering , Gitam University, India	Jun 2014 – Apr 2018

SKILLS

Programming and Database Managment: Python, SQL(MS SQL, MySQL), Snowflake, Microsoft SSMS

Statistical and Machine Learning Techniques: A/B testing, Hypothesis testing, Regression, Classification, Time-series Analysis, Clustering, CNNs, GANs, NLP

Data Visualization and Reporting: Tableau, Microsoft Power BI, Microsoft SSRS, MS Excel

Cloud Platforms and Big Data: AWS, Microsoft Azure, Hadoop, Spark

Data Processing and Analytics: Data Pipelines, Data Wrangling, Data Cleaning, Descriptive & Predictive Analytics

Tools and Version Control: Microsoft SSIS, Anaconda Navigator, Qt Designer, Jupyter Notebook, Git, GitHub, Trello, Agile, Waterfall, JIRA, Confluence

WORK EXPERIENCE

Senior Data Analyst, Arkos Health, Inc	Feb 2025 – Present
- Analyzed and synthesized healthcare data from EHR systems and cloud databases to deliver actionable insights that informed member engagement strategies, care coordination initiatives, and clinical performance monitoring across markets. - Designed and automated 15+ Tableau dashboards and Excel reports to track KPIs and regulatory metrics, reducing manual reporting time by 60% and enhancing real-time visibility for leadership, operations teams and business stakeholders. - Improved reporting accuracy by engineering robust QA processes, including custom validation logic and automated reconciliation across data pipelines—ensuring consistent and reliable insights. - Drove cross-functional collaboration with clinical, product, and analytics stakeholders to create and implement outreach strategies powered by analytics, resulting in a 45% lift in member engagement.	
Data Analyst, Cognizant Technology Solutions	Oct 2018 – Jun 2022
- Developed SQL-based business rules, Python automation scripts, and ETL workflows to streamline operations, achieving a 20% reduction in processing time and increased accuracy in reporting. - Created and monitored KPIs through Tableau dashboards, improving executive visibility into program performance and accelerating decision-making cycles by 25%. - Conducted A/B testing and statistical analyses on complex datasets to optimize user engagement strategies, contributing to a 15% boost in platform interaction.	

PROJECTS

UAV-based Street Road Inspection for Smart Cities Using Deep Learning (Demo)	Jan 2024 – Dec 2024
Objective: To Automate detection and severity classification of road anomalies to enhance transportation safety and infrastructure management. Implementation: Developed a modular two-stage framework integrating advanced deep learning techniques for precise anomaly detection and category-specific models for severity assessment, leveraging UAV imagery. Delivered actionable insights through an interactive dashboard for efficient decision-making. Tools: Python (PyTorch, OpenCV), Deep Learning (YOLOv8, MobileNetV1-SSD, Vision Transformers, Faster-RCNN)	
The Learning Agency Lab: PII-Data-Detection (Kaggle Competition) (GitHub)	Jan 2024 – May 2024
Objective: To develop a model that detects personally identifiable information (PII) in student writing. Implementation: Applied Natural Language Processing techniques to analyze and identify PII-related patterns, including names and email addresses. The model enhances data privacy by accurately detecting sensitive information, improving safety in learning environments. Tools: Python, Natural Language Processing (NLP) techniques, Deep Learning (DistilBERT)	
Object Detection Framework for Multi-Class Image Processing	Jan 2024 – May 2024
Objective: Build a robust object detection framework to accurately identify and localize multiple object classes in images. Implementation: Designed a custom Faster R-CNN model with a ResNet-50 backbone, leveraging advanced augmentation techniques and optimized training pipelines to enhance model generalization. Achieved a mean Average Precision (mAP) of 78%, delivering precise object detection across diverse categories. Tools: Python(PyTorch), Faster R-CNN, ResNet-50, Image Processing	
Toxic Comment Classification Using Deep Learning	Jan 2024 – May 2024

- **Objective:** To effectively detect toxic language and enhance content moderation for healthier online environments.
- **Implementation:** Leveraged preprocessing techniques like tokenization, stemming, and class balancing to enhance data quality. A CNN-LSTM hybrid model captured spatial and sequential text patterns, with accuracy improved through fine-tuning and optimization strategies like gradient clipping and learning rate scheduling across toxicity categories.
- **Tools:** Python(PyTorch, TorchText), NLP techniques, Neural Networks(CNN-LSTM)

AI-Driven Application for Diabetes Care Predictive ([GitHub](#))

Jan 2024 – May 2024

- **Objective:** To improve diabetes care by utilizing predictive analytics and offer personalized health recommendations.
- **Implementation:** Designed an intuitive, real-time application employing machine learning models to assess diabetes risk using individual health data. The system provides highly accurate predictions, enabling users to manage their condition with tailored lifestyle recommendations.
- **Tools:** Python (Pandas, Matplotlib, Seaborn, Streamlit), Machine Learning and Data Mining techniques

Natural Disaster Prediction Using Machine Learning ([GitHub](#))

Aug 2023 – Dec 2023

- **Objective:** Enhance global resilience by predicting future natural disasters and provide early preventative measures.
- **Implementation:** Developed and fine-tuned machine learning models, including classification and time-series analysis, to forecast natural disasters using historical data. Employed Python for data preprocessing and model training, ensuring high accuracy and reliability in predictions.
- **Tools:** Python (Pandas, Scikit-Learn, Matplotlib, Seaborn), Machine Learning (Classification, Time-Series Analysis)

Crime Prediction Using Machine Learning ([GitHub](#))

Aug 2023 – Dec 2023

- **Objective:** To create a predictive system for identifying high-risk areas and proposing data-driven strategies for safer neighborhoods.
- **Implementation:** Analyzed historical crime data to uncover patterns and develop predictive models using advanced data analysis techniques. Achieved a 91% F1-Score with tree-based models, enhancing resource allocation and strategic planning for crime prevention.
- **Tools:** Python (Pandas, Scikit-Learn, Matplotlib, Seaborn), Machine Learning(XGBoost, Random Forest, Decision Tree)

E-Commerce Application ([Demo](#))

Jan 2023 – May 2023

- **Objective:** Analyze marketing trends and revenue to optimize business performance and identify potential issues.
- **Implementation:** Built an integrated and interactive analytical application using Python (PyQt5) for real-time KPI monitoring and insights into customer behavior. Leveraged MySQL for data storage and retrieval while utilizing ETL processes with NumPy and Pandas to streamline data workflows.
- **Tools:** Python (PyQt5 GUI, NumPy, Pandas), MySQL, SQL Workbench

A Cross Country Region Wise Analysis of Adolescent Delinquency ([GitHub](#))

Jan 2023 – May 2023

- **Objective:** To understand the root causes of adolescent delinquency and identify potential interventions.
- **Implementation:** Conducted a comprehensive analysis of large-scale adolescent delinquency datasets, utilizing Tableau for data visualization and statistical analysis to uncover patterns, correlations, and regional hotspots of delinquent behavior. The findings were presented through interactive dashboards, enabling data-driven decisions for effective interventions.
- **Tools:** Tableau, Python (Pandas, NumPy, Seaborn, Matplotlib)